



Pizza Sales Performance Analysis Report

PIZZA SALES 2015

Power BI Final Test Report

Dataset: 2015 Pizza Sales Dataset

Analysis period: 2015

Prepared by: Van Anh Dinh

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1. Executive Summary

This report analyses a pizza sales transaction 2015 dataset using Power BI. The analysis focuses on sales performance, product and categories performance, time trends, ingredient analysis, and customer ordering behaviour.

- Total revenue was \$817.86K, which was from 21.35K distinct orders and 49.57K pizzas sold.
- Average Order Value was \$38.31, and the average number of pizzas purchased per order was 2.32.
- Classic was the best seller pizza category, generating \$220.05K revenue and 14,888 pizzas sold.
- Friday was the busiest day with 3,538 orders, while Sunday was the quietest day with 2,624 orders.
- Garlic and tomatoes were the highest volume ingredients.

2. Introduction

2.1. Business Context

The dataset has historical pizza sales data for 2015, but the raw dataset is a transactional file containing repeated product, category, size, price and ingredient information. In this form, the data is difficult for managers to use consistently for performance monitoring and operational decisions.

2.2. Objectives

The main objective of this project is to use Power BI to transform, model and analyse the 2015 pizza sales data and convert it into clear, decision-focused information.

- Clean, validate and restructure the source data using Power Query
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- Analyse daily, weekly and monthly sales trends and anomalies.
- Identify peak weekdays and hours based on Total Orders.
- Compare pizza categories, pizza names and sizes by sales performance.
- Analyse ingredient popularity and the relationship between average price and sales volume.
- Provide practical recommendations to improve sales performance.

2.3. Business Problem

Managers need to understand not only how much the company sold, but also when demand is highest, which products boost performance, which ingredients are important, and how customers structure their orders.

2.4. Business Questions

- (1) What was total revenue the company generated in 2015?
- (2) What was the average revenue per order and average number of pizzas per order?
- (3) How did sales change by day, week and month?
- (4) Which weekdays and hours had the highest and lowest order volume?
- (5) Which dates experienced significant spikes or drops?
- (6) Which pizza categories performed best by quantity sold and revenue?
- (7) Which pizza names were the top and bottom revenue or quantity?
- (8) Which sizes were preferred within each category?
- (9) Which ingredients were most and least popular?
- (10) How common were single item versus multi item orders?

3. Data Preparation and Modelling

3.1. Dataset Tables and Data Quality Check

Table 1: Overview of Source Dataset Tables

Table	Role	Meaning
data_pizza.csv	CSV file	Contains 2015 order ID, date, time, pizza ID, quantity, size, price, name, category and ingredient data.
data_dictionary.xlsx	Supporting reference	Describes the fields contained in the transaction dataset.

Data quality checking was completed in Power Query using column quality, column distribution, and column profile. No major missing value, error value, or data type issue was found. Text fields were trimmed and cleaned as a preventative quality step, no material inconsistencies were observed.

Table 2: Data Quality Assessment Summary

Check	Result	Interpretation
Validity	100% valid values reported	The dataset is suitable for analysis.
Errors	No major error values found	No major data cleaning issue.
Empty	0% empty reported	The dataset is suitable for analysis.
Data types	Corrected or checked before modelling	Fields can be used correctly in measures and visuals.

3.2. Analytical Columns Created

Table 3: Calculated Columns

Column	Logic / Definition	Purpose
size_sort	S = 1, M = 2, L = 3, XL = 4, XXL = 5	Ensures pizza sizes are displayed in the correct logical order.
order_quantity	Total quantity grouped by order_id	Calculates the total number of pizzas included in each distinct order.
order_type	Order Quantity = 1: single Order Quantity > 1: multi	Classifies each order to compare the frequency of single and multi item orders.

3.3. Star Schema Data Model

The transformation process was designed to reduce repeated descriptive values and support stable relationships in the model.

The FactSales table remains at one pizza line item within an order. Descriptive fields are stored in dimensions, while transaction identifiers, quantities, dates, times and foreign keys are retained at fact level.

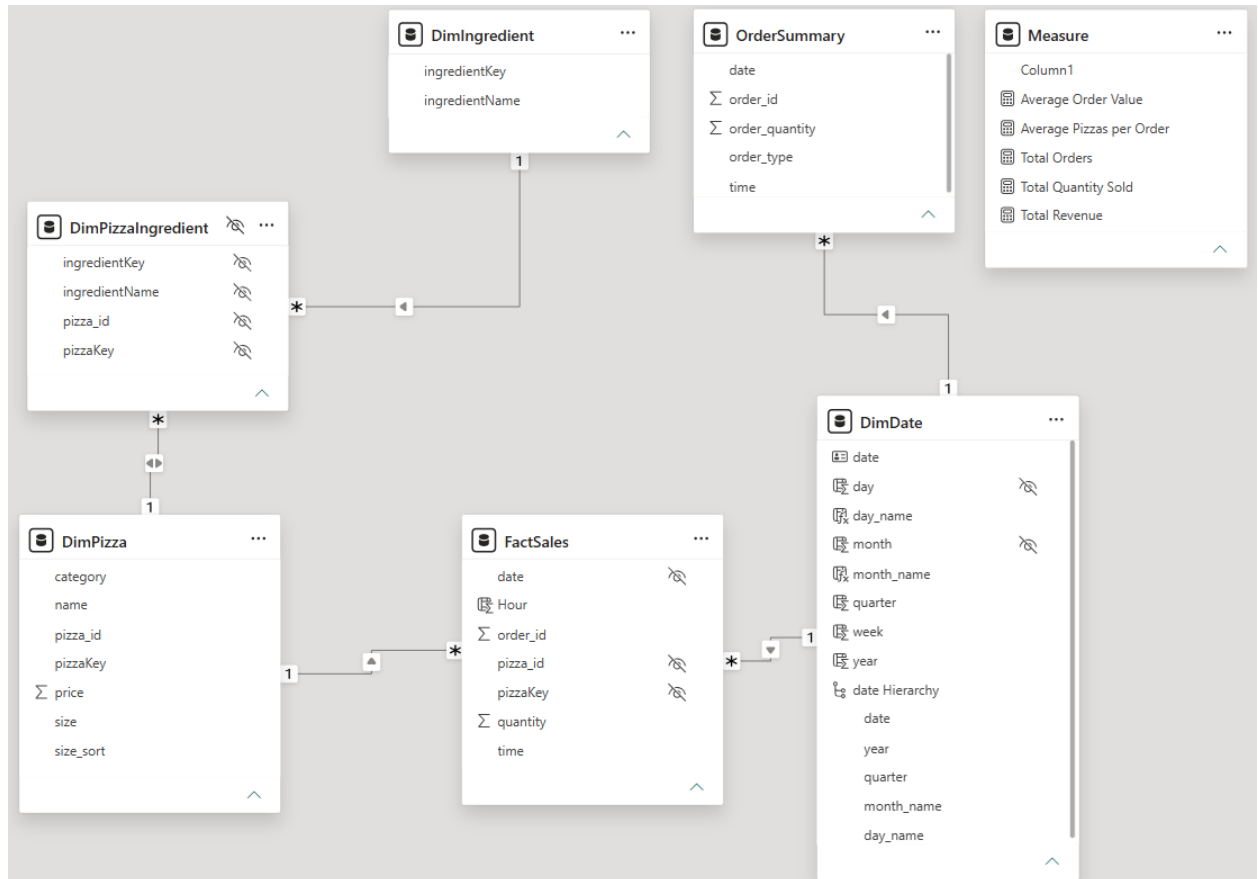


Figure 1: Power BI Star Schema model design

3.4. Key DAX Measures

Table 4: Core DAX Measures

Measure	DAX / Definition	Purpose
Total Revenue	SUMX(FactSales, FactSales[quantity] * RELATED(DimPizza[price]))	Total sales revenue generated.
Total Quantity Sold	SUM(FactSales[quantity])	Total number of pizzas sold.
Total Orders	DISTINCTCOUNT(FactSales[order_id])	Number of distinct customer orders.
Average Order Value	DIVIDE([Total Revenue], [Total Orders])	Average revenue generated per order.
Average Pizzas per Order	DIVIDE([Total Quantity Sold], [Total Orders])	Average pizza quantity in each order.
Average Pizza Price	AVERAGE(DimPizza[price])	Average listed price in the current filter context.

4. Dashboard Visualisation Design

There are four interactive dashboard was developed in Power BI. The dashboard follows a progressive flow from overall performance to detailed analysis: Executive Overview, Sales Trends, Product Analysis, and Ingredient & Order Analysis. A navigation page and home buttons were used to improve user experience.

Table 5: Dashboard Structure and Analytical Purpose

Dashboard page	Main purpose	Main visuals
Executive Overview	Provide high level business 2015 performance summary.	Five KPI cards, monthly revenue trend, revenue share by category, slicers.
Sales Trends	Analyse temporal patterns, weekday and hour demand and anomalies.	Daily anomaly trend, weekday trend, weekly trend, weekday and hour heatmap.
Product Analysis	Compare category, pizza_name and size performance.	Category ranking table, top/bottom pizza charts, category and size distribution, revenue/quantity ranking.
Ingredient & Order Analysis	Investigate ingredients, price volume patterns and ordering behaviour.	Ingredient rankings, price volume scatter, pizzas per order, single/multi order analysis.

Table 6: Dashboard Visuals and Supporting Measures

Visual / Table	Fields / Measures used	How it supports analysis
Cards	Total Revenue, Total Orders, Total Quantity Sold, Average Order Value, Average Pizzas per Order	
Monthly Revenue Trend	month_name, Total Revenue	Shows monthly revenue movement across 2015
Revenue Share by Pizza Category	category, Total Revenue	Shows each pizza category's contribution to total revenue
Daily Revenue Trend	day_name, Total Revenue	Compares revenue performance daily
Weekly Revenue Trend	week; Total Revenue	Shows weekly changes in revenue throughout 2015
Revenue Trend and Anomalies	date, Total Revenue, anomaly detection	Identifies unusual sales spikes and drops on specific dates
Orders by Day and Hour Heatmap	day_name, Hour, Total Orders	Identifies peak and quiet ordering periods by weekday and hour
Pizza Category Ranking	category, Total Quantity Sold, Total Revenue	Compares category performance by sales volume and revenue

Visual / Table	Fields / Measures used	How it supports analysis
Top 5 Pizza Analysis	name, Total Quantity Sold / Total Revenue, Top 5 ranking	Identifies the highest performing individual pizzas
Bottom 5 Pizza Analysis	name, Total Quantity Sold / Total Revenue, Bottom 5 ranking	Identifies the lowest performing individual pizzas
Pizza Size Distribution	category, size, Total Quantity Sold	Compares pizza size preferences within each category.
Top 5 Ingredient Analysis	ingredient_name, Total Quantity Sold, Top 5 ranking	Identifies ingredients associated with the highest sales volume.
Bottom 5 Ingredient Analysis	ingredient_name, Total Quantity Sold, Bottom 5 ranking	Identifies ingredients associated with the lowest sales volume.
Price and Sales Volume	category, Average Pizza Price; Total Quantity Sold	Examines the relationship between average pizza price and sales volume across categories.
Single vs. Multi item Orders	Order_type; order_id (count)	Compares the frequency of single item and multi item orders.

5. Data Analysis and Findings

5.1 Executive Overview

The Executive Overview shows that the business generated approximately \$817.86K in revenue from 21.35K distinct orders during 2015. A total of 49.57K pizzas were sold. The Average Order Value was \$38.31 and customers purchased an average of 2.32 pizzas per order.

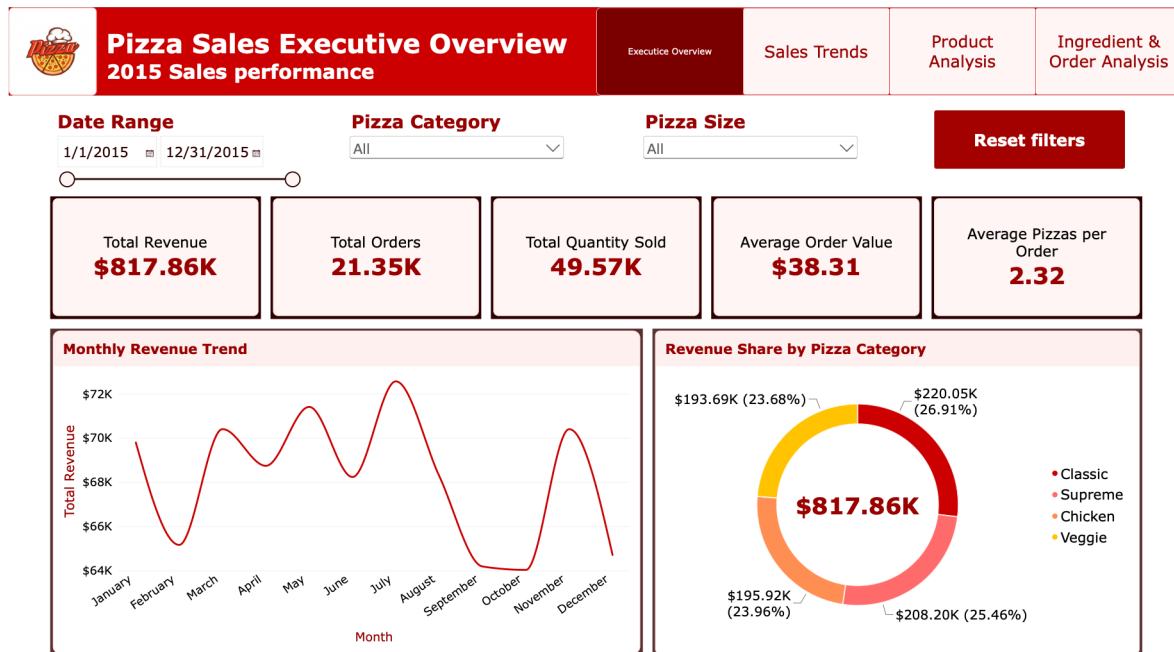


Figure 2: Executive Overview dashboard screenshot

Business Question 1: What was the total revenue generated by the company in 2015?

Business Question 2: What were the average revenue per order and the average number of pizzas per order?

Table 7: Overall Sales Performance

KPI	Results	Interpretation
Total Revenue	\$817.86K	Overall annual sales value
Total Orders	21.35K	Distinct orders recorded during the year
Total Quantity Sold	49.57K	Total pizzas sold
Average Order Value	\$38.31	Average revenue per order
Average Pizzas per Order	2.32	Average quantity of pizzas in an order

The category revenue share is relatively balanced, but Classic is the largest share at \$220.05K (26.91%). Supreme contributes \$208.20K (25.46%), Chicken \$195.92K (23.96%), and Veggie \$193.69K (23.68%). This indicates that no single category completely dominates revenue, although Classic is the strongest contributor.

5.2 Sales Trends and Anomalies

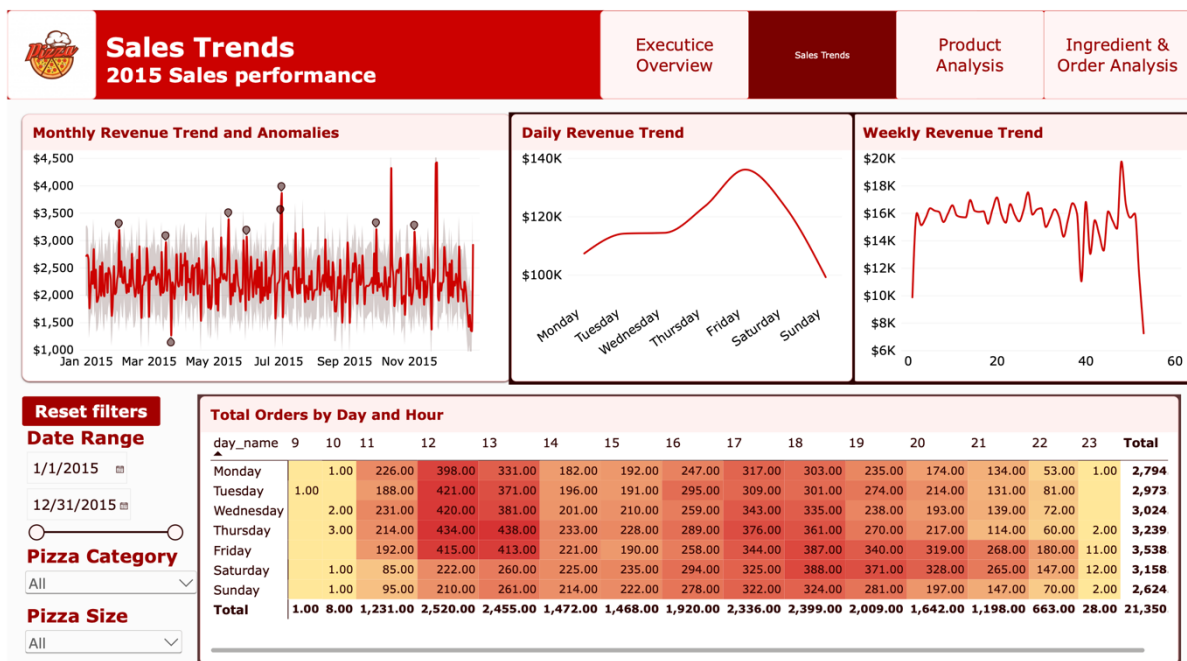


Figure 3: Sales Trends dashboard screenshot

Business question 3: How did sales change by day, week and month throughout 2015?

Table 8: Sale Trends

Period	View	Interpretation
Monthly	Revenue fluctuated, with July showing the strongest monthly level and weaker performance around September and October.	Sales were not evenly distributed across the year.
Weekly	Weekly revenue remained variable, with several noticeable peaks and drops across the year.	Short term sales demand changed from week to week.
Daily	Revenue generally increased toward Friday before falling over the weekend, with Sunday showing the weakest daily level.	Sales performance varied considerably across weekdays.

Business question 4: Which weekdays and hours recorded the highest and lowest order volumes?

The **daily analysis** shows that Friday recorded the highest order volume with 3,538 orders. Thursday was also strong with 3,239 orders, followed by Saturday with 3,158 orders. In contrast, Sunday recorded the lowest order volume at 2,624 orders, while Monday was the second lowest at 2,794 orders.

This pattern shows that customer demand generally becomes stronger toward the end of the working week. Friday is therefore the most important day in terms of order volume, while Sunday experiences comparatively lower demand.

The **hourly analysis** identifies two clear periods of high order activity. The strongest period occurs during lunch, particularly at 12:00 with 2,520 orders, which is the highest hourly

total in the dataset. Demand remains high at 13:00 with 2,455 orders. A second peak occurs during the evening. Order volume rises from 16:00 and reaches 2,336 orders at 17:00 and 2,399 orders at 18:00, before gradually declining after 19:00.

The main insight is that customer ordering activity is concentrated around lunch and dinner periods.

5.3 Product Analysis

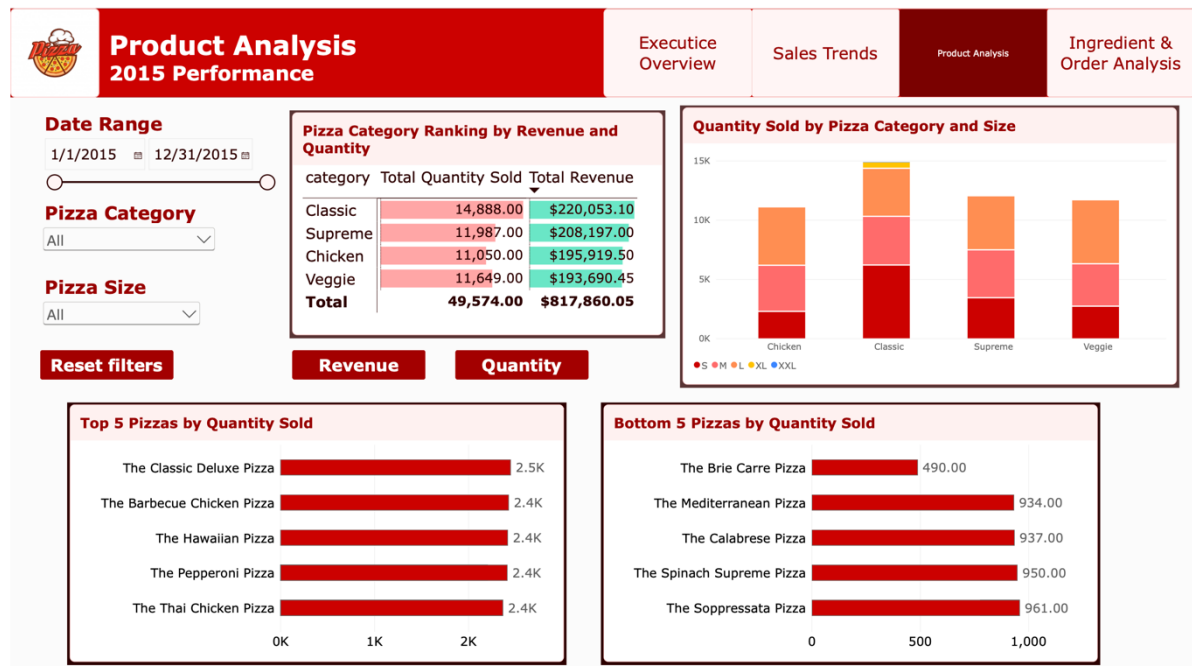


Figure 4: Product Analysis dashboard screenshot

Business question 6: Which pizza categories performed best by quantity sold and revenue?

Table 9: Pizza Category Performance

Category	Total Quantity Sold	Total revenue	Interpretation
Classic	14,888	\$220,053.10	Strongest category by both quantity sold and revenue.
Supreme	11,987	\$208,197.00	Second-highest category by both measures.
Chicken	11,050	\$195,919.50	Lowest quantity sold but third-highest revenue.
Veggie	11,649	\$193,690.45	Higher quantity than Chicken but lowest revenue.

The key insight is that product category performance should not be judged using quantity sold alone. Revenue and sales volume should be considered together because a category can generate comparatively stronger revenue even when its quantity sold is lower.

Business question 7: Which pizza names were the top and bottom performers by quantity sold?

The **top product analysis** shows that The Classic Deluxe Pizza recorded the highest quantity sold at approximately 2.5K pizzas. The Barbecue Chicken Pizza, Hawaiian Pizza, Pepperoni Pizza and Thai Chicken Pizza each recorded approximately 2.4K pizzas sold, placing them among the strongest-performing individual products.

At the opposite end of the ranking, The Brie Carre Pizza recorded only 490 pizzas sold, making it the lowest-performing pizza by quantity in the displayed analysis. The Mediterranean Pizza, Calabrese Pizza, Spinach Supreme Pizza and Soppresata Pizza also sold fewer than 1,000 units each.

The Product Analysis dashboard includes **Revenue and Quantity buttons**, allowing the same top and bottom analysis to be viewed using either measure.

Business question 8: Which pizza sizes were preferred within each category?

The main insight is that Large appears to be the preferred pizza size across categories. This information is useful for menu and inventory planning because it indicates which size options contribute most strongly to sales volume.

5.4 Ingredient and Order Analysis

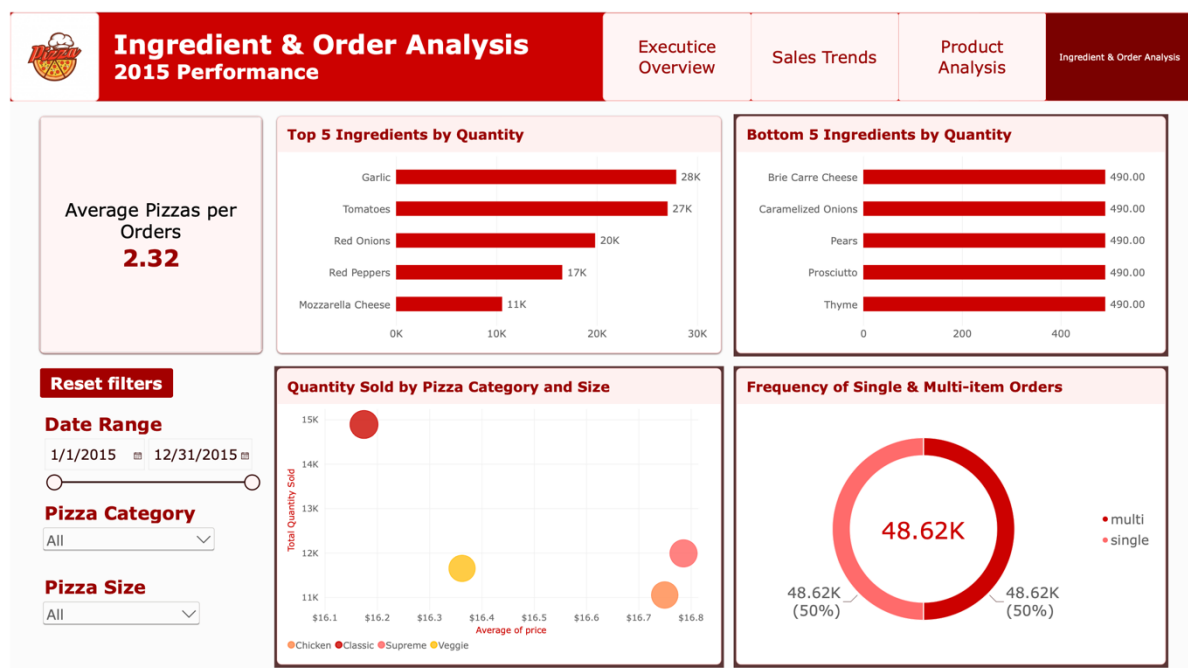


Figure 5: Ingredient & Order Analysis dashboard screenshot

Business question 9: Which ingredients were the most and least popular based on quantity sold?

The ingredient analysis shows that **Garlic is the most popular ingredient**, associated with approximately 28K pizzas sold. Tomatoes follow closely at approximately 27K, while Red Onions, Red Peppers and Mozzarella Cheese complete the top five. These ingredients appear across pizzas with substantial sales quantities, making them important from an inventory perspective.

At the lower end of the ranking, **Brie Carre Cheese, Caramelized Onions, Pears, Prosciutto and Thyme** each show a quantity of 490. These ingredients therefore appear in pizzas with substantially lower sales volume than the top ingredients.

The key business insight is that frequently used ingredients such as Garlic and Tomatoes are likely to require greater attention in stock planning because they are associated with much higher sales volumes. Less frequently used ingredients should also be monitored because their lower usage may affect inventory efficiency.

Business question 10: How common were single item orders compared with multi item orders?

The current dashboard donut chart displays 48.62K for both Single and Multi items. However, this result does not align with the Executive Overview, which reports only 21,350 distinct orders. Therefore, the current donut result should not yet be interpreted as customer ordering behaviour. The order level calculation needs to be validated so that the chart counts distinct orders rather than transaction rows.

6. Recommendations

Table 10: Strategic Recommendations Based on Key Findings

Key issue identified	Deeper business insight	Recommended action	Expected benefit
Friday and lunch/dinner periods have the highest order volume	Demand is concentrated in predictable weekly and hourly period	Increase staffing and ingredient readiness before Friday lunch and evening peaks	Reduces peak period pressure and supports faster order handling
Classic leads quantity and revenue	Classic products are central to both sales volume and revenue contribution	Protect availability of Classic pizzas and prioritise their high-use inputs in inventory planning	Reduces stock out risk in the strongest category
Chicken earns more revenue than Veggie despite lower quantity	Unit volume alone does not explain category value	Evaluate revenue, quantity and average price together when making menu decisions	Avoids removing products that generate strong value per unit
Large pizzas contribute a substantial share of volume	Size demand affects packaging, dough and ingredient consumption	Prioritise operational capacity for high volume sizes and review whether low volume XL/XXL variants justify complexity	Improves production and packaging planning
Garlic and tomatoes are high volume ingredients	Shortages of common ingredients could affect multiple popular pizzas at the same time	Maintain appropriate safety stock and supplier reliability for highly used ingredients	Improves menu availability and inventory resilience
Several pizza names sell below 1,000 items	Low volume menu items create complexity but may still have revenue or niche value	Review weak products using revenue, ingredient cost and strategic value before promoting, redesigning or removing them	Reduces unnecessary menu complexity while protecting valuable niche items
September and October are visually weaker months	Sales are not evenly distributed across the year	Investigate weaker periods and test targeted promotions or campaigns rather than applying broad discounts	Supports more stable demand without unnecessary price reduction
Price and volume relationship is weak	The current category level scatter does not support a simple price demand conclusion	Test pricing decisions at pizza and size level and monitor both revenue and quantity	Reduces the risk of unsupported pricing changes

Key issue identified	Deeper business insight	Recommended action	Expected benefit
Single vs multi item result is currently incorrect	Transaction row counts are being displayed instead of distinct order classification	Correct the OrderSummary logic before using the result for bundle or basket-size decisions	Ensures recommendations are based on valid order level evidence

7. Limitations

This report has several limitations. First, the analysis is mainly descriptive, meaning it identifies patterns in revenue, quantity, order timing, products, sizes and ingredients, but it cannot fully explain the causes behind those patterns.

Second, the dataset does not provide product cost or profit, customer demographics, store locations, promotion history, inventory levels, staffing levels, preparation completion timestamps. As a result, the report can analyse revenue but not profitability, cannot identify customer segments, and cannot confirm that promotions or operational events caused sales anomalies.

Finally, some insights are based on aggregated dashboard views. Managers should drill into product, size, hour and individual dates before implementing major menu, pricing or staffing modification.

8. Conclusion

The Power BI project converts a raw 2015 pizza transaction dataset into a structured analytical model and four pages interactive dashboard. Overall performance was strong enough to generate \$817.86K in revenue from 21.35K orders and 49.57K pizzas sold, with an Average Order Value of \$38.31 and an average of 2.32 pizzas per order.

Classic was the strongest category by revenue and quantity. Demand was highest on Friday and concentrated around lunch and early evening, providing useful evidence for staffing and preparation planning. Product analysis identified a clear gap between the strongest and weakest pizza types, while size and ingredient analysis highlighted operational priorities for production and stock management.

Overall, the dashboard provides a practical decision support tool for sales monitoring, menu planning, inventory preparation and operational scheduling. Future analysis would be strengthened by adding cost, promotion, customer, and service time data so that profitability can be examined in greater depth.

Reference

Use of Generative AI: ChatGPT was used as a supporting tool to provide suggestions on dashboard formatting, assist in developing the approach for analysing single item versus multi item order frequency, and provide feedback on the structure, clarity, and presentation of the report.